

GitHub Project Repository Setup Guide

Purpose: This guide will help you organize your project using a clean, professional GitHub structure. This structure ensures your work is easy to follow, reproducible, and ready for presentation.

Step 1: Create Your GitHub Repository

1. Go to GitHub and click **“New Repository”**
2. Name your repository (use a clear project name)
3. Set the repository to **Public**
 - Use public unless your instructor explicitly tells you otherwise
4. Initialize with a **README file**
5. Click **Create Repository**

Step 2: Add me (your instructor)

1. Go to your repository
2. Click **Settings**
3. Click **Collaborators**
4. Click **Add people**
5. Enter maisAlraee
6. Send the invitation

Step 3: Create the Required Folder Structure

Inside your repository, create the following folders:

note: GitHub doesn't let you create empty folders, so you'll create a file inside each folder to generate it.

Let us do one together: **Create data/raw**

1. Click **Add file** → **Create new file**
2. In filename box, type: data/raw/.gitkeep

Git (what GitHub uses) does NOT track empty folders. If a folder has no files inside, Git will ignore it, and it won't show up in your repo

Now Repeat the same process for:

data/interim/.gitkeep
data/processed/.gitkeep
data/external/.gitkeep

...

The structure should look like:

```
data/
├── raw/
└── processed/
```

```
ingestion/
├── notebooks/
└── docs/
```

```
eda/
├── notebooks/
├── docs/
└── images/
```

```
analysis/
├── notebooks/
├── docs/
└── images/
```

```
models/
```

```
reports/
├── plan/
├── final/
└── figures/
```

```
docs/
├── references/
└── sources/
```

Step 4: Understand What Goes Where

Data Folder

- **raw/** → original data (do not modify)
- **processed/** → final datasets used for analysis

Ingestion Folder

- Loading and cleaning data
- Example:
 - Reading CSV files/Fixing missing values

EDA (Exploratory Data Analysis)

- Graphs, summaries, early insights
- Example:
 - Histograms/Correlation plots

Analysis Folder

- Final modeling and results
- Example:
 - Statistical models/Machine learning

Models Folder

- Saved models or outputs

Reports Folder

- Presentations and final deliverables
- Includes:
 - Project plan
 - Final report

Step 5: Where Your Code Goes (IMPORTANT)

Put your .qmd , .Rmd or .ipynb files in:

ingestion/notebooks/
eda/notebooks/
analysis/notebooks/

Scripts (Reusable Code)

Put .R or .py files in: scripts/ [inside the appropriate section (ingestion, eda, or analysis)]

Step 6: Naming Convention

Use clear, consistent file names:

1.0-initial-data-cleaning.qmd
1.1-eda-summary.ipynb
2.0-final-model.R

Format: [number]-[short-description]

Best Practices

- Do NOT modify files in data/raw/
- Keep your folders organized at all times
- Write clear file names (no random names like “final_final_v2”)
- Add comments in your code
- Push your work regularly (you will learn this next)

Final Checklist

Before moving on, make sure:

- Repository is **public**
 - I am added as a collaborator
 - Folder structure is created
 - You understand where your files go
 - You are ready to start adding code
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